FORMAL FUNCTIONS FOR PROPER REPRESENTABLE MAPS OF NOETHERIAN SPACES: AN ISOMORPHISM OF COMPLETED PULLBACKS YIELDS AN ALMOST-ISOMORPHISM ON INVERSE SYSTEMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9940-9960 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 9962-9987, the complete author proof (24 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Companion to the abelianness card L16066; the kernel/cokernel torsion is measured by powers of K, the ideal of the non-isomorphism locus complement.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Proper pushforward of coherent sheaves on spaces (Cohomology of Spaces, lemma-proper-pushforward-coherent), the schemes formal functions theorem (Cohomology of Schemes, lemma-inverse-systems-push-pull), and the etale-descent formalism for Coh(X, I) set up in the chapter's preceding section.

Source reference. Original lemma 088B: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9987–9987. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08AR: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9987–9987. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a representable proper morphism of Noetherian algebraic spaces over S. Let $\mathcal{J}, \mathcal{K} \subset \mathcal{O}_Y$ be quasi-coherent sheaves of ideals. Assume f is an isomorphism over $V = Y \setminus V(\mathcal{K})$. Set $\mathcal{I} = f^{-1}\mathcal{J}\mathcal{O}_X$. Let $(\mathcal{G}_n)$ be an object of Coh$(Y, \mathcal{J})$, let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module, and let $\beta : (f^*\mathcal{G}_n) \to \mathcal{F}^\wedge$ be an isomorphism in Coh$(X, \mathcal{I})$. Then there exists a map*

$$\alpha : (\mathcal{G}_n) \longrightarrow (f_*\mathcal{F})^\wedge$$

in Coh$(Y, \mathcal{J})$ whose kernel and cokernel are annihilated by a power of $\mathcal{K}$.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Since f is a proper morphism we see that $f_*\mathcal{F}$ is a coherent $\mathcal{O}_Y$-module (Cohomology of Spaces, Lemma 08AR). Thus the statement of the lemma makes sense. Consider the compositions

$$\gamma_n : \mathcal{G}_n \to f_*f^*\mathcal{G}_n \to f_*(\mathcal{F}/\mathcal{I}^n\mathcal{F}).$$

Here the first map is the adjunction map and the second is $f_*\beta_n$. We claim that there exists a unique α as in the lemma such that the compositions

$$\mathcal{G}_n \xrightarrow{\alpha_n} f_*\mathcal{F}/\mathcal{J}^n f_*\mathcal{F} \to f_*(\mathcal{F}/\mathcal{I}^n\mathcal{F})$$

equal γ_n for all n. Because of the uniqueness and étale descent for $Coh(Y, \mathcal{J})$ it suffices to prove this étale locally on Y. Thus we may assume Y is the spectrum of a Noetherian ring. As f is representable we see that X is a scheme as well. Thus we reduce to the case of schemes, see proof of Cohomology of Schemes, Lemma 088B. □